

1 Chapter 1: Signals and Signals

In general, signals refer to sets of data that vary as a function of time (or some other independent variable), while systems process signals (for some given input, the systems returns a predictable output).

1.1 Size of a Signal

It's often useful to characterize a signal, which can often be mathematically complex, into a more manageable quantity, which we will refer to here as its size. Further, a signal's size can be expressed in terms of energy or power (don't confuse this with the more common definition of power as joules per second, or watts).

We'll begin with energy. Mathematically, a signal $x(t)$'s energy is defined as

$$E_x = \int_{-\infty}^{\infty} |x(t)|^2 dt$$

and defines the signal's integral (area under the curve) over an infinite length of time. Obviously, the result is only significant if the signal's energy is *finite*. For example, a 1-second unit impulse $x(t) = (u(t) - u(t - 1))$ would have a finite energy $E_x = 1$ (its definite integral from $-\infty$ to *infy*), while a sinusoidal function $g(t) = \sin(t)$ would have an infinite energy $E_g = \infty$, due to its periodic nature.

When a signal's energy is infinite, we can use power as a more useful measure of the signal's size. Power is defined mathematically as

$$P_x = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} |x(t)|^2 dt$$

and defines the signal's average energy. On the other hand, power is not a useful way to measure the size of a signal with finite energy. For example, a sinusoidal function $x(t) = \sin(t)$ would have a power of $P_x = \frac{1}{2}$, while the unit function given before, $g(t) = u(t) - u(t - 1)$, would have a power of $P_g = \frac{1}{\infty} = 0$.

The above formulas can be used to define two types of signals: energy signals and power signals. Energy signals are those with finite energy, and power signals are those with finite **and** non-zero power.

1.2 Useful Signal Operations

Common signal operations include time shifting, scaling, and reversal.

Time shifting refers to the displacement of a signal in time, without changing any of its other properties. The simplest way to accomplish this is by replacing all instances of t with $t - T$, such that $\phi(t) = x(t - T)$ (the sign of T determines whether the signal shifts to the right, delaying it, or to the left, advancing it.)

Time scaling refers to the compression or expansion of a signal in time, and can be accomplished by replacing all instances of t with at , where a represents

the compression/expansion factor, such that $\phi(t) = x(at)$ (the value of a will determine whether the signal is expanded or compressed in time.)

Finally, time reversal "flips" the signal in time, and can be accomplished by replacing all instances of t with $-t$, such that $\phi(t) = x(-t)$.

1.3 Classification of Signals

Signals can be classified by a number of their characteristics, as follows:

1. Continuous versus discrete time signals:

Continuous time signals have a value $x(t)$ for every t in the domain, while discrete time signals only have values $x(t)$ for discrete values of t (usually given in equally spaced intervals, though not always).

2. Analog versus digital signals:

Analog signals may take on any value in $(-\infty, \infty)$; that is, their range is infinite, while digital signals have a range limited to a finite range (the most familiar example is binary, where $x(t)$ can take on one of two discrete values).

3. Periodic versus aperiodic signals:

Periodic signals take on the form $x(t) = x(t \pm aT_0)$, implying that over some range a , the signal repeats itself. This is not the case with aperiodic signals, whose behavior is less predictable but not entirely unpredictable (more on that later).

4. Causal versus non-causal signals:

Causal signals begin at $t = 0$, before which they are not defined, while non-causal signals are defined prior to $t = 0$ (and after).

5. Deterministic versus probabilistic (random) signals:

Deterministic signals are described for every point in their domain, usually by a mathematical function, while probabilistic or random signals are not defined specifically, and are only known in terms of other characteristics, such as their mean value, mean-squared value, et cetera.

1.4 Some Useful Signal Models

One of the most useful signal models is the unit step, defined by

$$u(t) = \begin{cases} 1 & t > 0 \\ 0 & t \leq 0 \end{cases}.$$

When describing piecewise functions, the unit step may be used to define each condition of the piecewise function. For example, when defining a triangular-shaped that is 2 seconds long and begins at $t = 0$, we may use a unit step

function of width 1 $u(t) - u(t - 1)$ to define the first half of the triangle, and a second, right-shifted unit step function of width 1 $u(t - 1) - u(t - 2)$ to define the second half. In mathematical terms,

$$x(t) = \begin{cases} t & 0 \leq t \leq 1 \\ -t + 2 & 1 \leq t \leq 2 \end{cases} = [u(t) - u(t - 1)]t + [u(t - 1) - u(t - 2)](-t + 2).$$

Next, the unit impulse (or delta dirac) function $\delta(t)$ is defined mathematically in terms of its integral, such that

$$\int_{-\infty}^{\infty} \delta(t) dt = 1; \delta(t) = 0, t \neq 0.$$

The above implies that at $t = 0$, $\delta(0) = \infty$, since it is zero-valued for all other values of t and yet has a definite integral of 1.

As a result, the unit impulse obeys the **sampling** or sifting property, whereby its integral picks out the value of an associated function at $t = 0$ (or wherever $\delta(t)$ is time-shifted to). For example, consider a function $\phi(t) = 2t + 1$. When integrated and multiplied by the delta function, it follows that

$$\int_{-\infty}^{\infty} \phi(t) \delta(t) dt = 1 \int_{-\infty}^{\infty} \delta(t) dt = \phi(T) = 1.$$

We can conclude based on the above that the delta function, when used in this way, will "sample" the value of $\phi(t)$ at the point T , given the delta function is shifted; e.g., $\delta(t - T)$.

1.5 Classification of Systems

Systems may be expressed mathematically as a function of the form

$$y(t) = x(t),$$

where $x(t)$ is the input to the system and $y(t)$ represents the output of the system.

Systems may be categorized as linear and non-linear, with this class limiting itself to the analysis of linear systems. Linearity is defined by two separate properties: *additivity* and *homogeneity*. The additivity property says that if some inputs x_1 and x_2 are applied to the system, resulting in outputs y_1 and y_2 , respectively, then the sum of those inputs, $x_1 + x_2$, when applied to the system, will result in an output $y_1 + y_2$; that is, the sum of their respective outputs when applied independently. Homogeneity, the second property of linear systems, says that if the input to a system x_1 is increased by a factor of k so that the input is equal to kx_1 , the output will increase by the same factor, ky_1 .

In combination, the two properties yield in a linear system, one in which $k_1x_1 + k_2x_2 = k_1y_1 + k_2y_2$, a property called *superposition*.

Systems may further be categorized as time-invariant and time-variant systems. Put simply, a time-invariant system whose input is time-shifted by some

arbitrary constant τ_0 will result in an equivalently time-shifted output τ_0 , while a time-variant system's output *will not* obey this relationship. Mathematically, given some input $x_1(t)$ and the matching output $y(t)$, if a time-shifted input $x_2 = x_1(t \pm \tau_0)$ is applied to the system, the output should equal $y_2(t) = y_1(t \pm \tau_0)$ in a time-invariant system. By applying a time shifted input to the system and comparing the corresponding, time shifted output, we can determine whether a system is time-invariant or time-variant.

Systemes may also be categorized as instantaneous (memoryless) or dynamic systems; the output of dynamic systems depends only on the input at the same instant, whereas dynamic systems may depend on inputs in the past or future. A subset of dynamic systems are those which are causal or non-causal; the output of causal systems depend only on inputs in the present or past, while noncausal systems' outputs depend on *future* values of the input.

Finally, invertible and non-invertible systems are categorized on the basis of whether the exact input can be recovered by manipulating the output. For example, the system $y = 2x$ is considered invertible, since by reversing the operation, $\frac{y}{2} = x$, the original input can be recovered. On the other hand, $y = x^2$ is not an invertible system, since the inverse operation $\sqrt{y} = x$ may yield a positive or negative value, the original input may not be recovered.